MERN Assignment Support

Implementation Overview · Web Evaluator Project · Pankaj Sahu

Overview

Platform ki existing architecture ko preserve karte hue **MERN assignment support** add kiya gaya hai — bina tech stack change kiye.

Implementation Details

1 Problem

Platform sirf HTML/CSS/JS assignments run kar sakta tha, kyunki student ka code ek sandboxed iframe mein chalta hai jahan Node.js, Express, ya MongoDB chal hi nahi sakte. MERN assignments possible nahi the.

2 Solution

Ek **`project_type`** column add kiya `projects` table mein jo 3 values le sakta hai:

- **live_preview** — purane HTML/CSS/JS assignments ke liye (default)
- **code_review** — MERN code likhwane ke liye (Monaco editor, no execution)
- **external_link** — full MERN projects ke liye (GitHub + deployed URL submit)

Platform type ke basis par UI switch kar deta hai.

3 Implementation Approach

Teen main changes kiye:

- Supabase schema mein **1 column** projects pe, **4 columns** submissions pe add kiye — sab backwards compatible with defaults.
- React side pe admin aur student dono pages mein ek simple `switch(project_type)` lagaya jo correct component render karta hai.
- Do naye components banaye — **CodeReviewMode.tsx** aur **ExternalLinkMode.tsx** — jo apne-apne flow handle karte hain.

4 Safety & Impact

Purane assignments ko kuch nahi hua — default **live_preview** ban gaye, iframe sandbox untouched hai. RLS policies change nahi karni padi. Teammate ki **ap-1** branch se bhi conflict nahi.

Frequently Asked Questions

Design aur implementation ke technical decisions ka rationale:

Question	Answer
Code run kaise hoga MERN ka?	Platform code execute nahi karega — ya toh instructor manually review karega **code_review** mode mein, ya student khud deploy karke link submit karega **external_link** mode mein. Ye industry-standard approach hai.
RLS policies kyun nahi badli?	Kyunki existing policies submission ki shape nahi dekhti — sirf assignment ka time window aur ownership check karti hain, jo dono modes mein same hain.
MongoDB kyun nahi use kiya?	Platform Supabase (Postgres) pe hai. MongoDB add karne se backend split ho jata, infra complex hoti. Jo add kiya hai usse MERN assess karne ka purpose solve ho jata hai bina stack tooda.
ZIP upload kyun nahi?	V2 mein consider karenge. V1 mein GitHub link + deployed URL kaafi hai, aur real-world mein engineers aise hi code review karte hain.
Testing kaise kiya?	Teenon types ka ek-ek project banake test student se end-to-end flow verify kiya — assignment creation, student submission, admin review.

Summary

Existing platform mein ek project type system add kiya — teen modes support karta hai: live preview (purana), code review (MERN code), aur external link (full MERN projects). Database mein 1 column add hua, UI type ke hisaab se branch karti hai, aur backwards fully compatible hai.

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